BEFORE YOU READ

Ranges of the common families

Knowing the family tells you the range before any calculation. Here is the range of each common function type.

- L0 **Range by function type**
  - L1 **Polynomial and absolute value**
    - L2 **Constant** f(x) = c — $\{c\}$
    - L2 **Linear** mx + b, m ≠ 0 — $(-\infty, \infty)$
    - L2 **Quadratic, a > 0** vertex at height k — $[k, \infty)$
    - L2 **Quadratic, a < 0** vertex at height k — $(-\infty, k]$
    - L2 **Absolute value** $|x|$ — $[0, \infty)$
  - L1 **Roots**
    - L2 **Square root** $\sqrt{x}$ — $[0, \infty)$
    - L2 **Cube root** $\sqrt[3]{x}$ — $(-\infty, \infty)$
  - L1 **Exponential and logarithmic**
    - L2 **Exponential** a^x — $(0, \infty)$
    - L2 **Logarithmic** $\log_a x$ — $(-\infty, \infty)$
  - L1 **Trigonometric**
    - L2 **Sine and cosine** — $[-1, 1]$
    - L2 **Tangent** — $(-\infty, \infty)$

TAKEAWAY

Recognise the family first

Identify the family, read off its basic range, then adjust for any **transformations**.